

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 2nd Term Examination-2022
LE 3219
Leather Microbiology

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define bacteria. State the chemical composition and function of the following components of a bacterial cell: (i) Cell wall (ii) Cell Membrane (iii) Nucleoid 10
- 1(b) Explain why strict anaerobes cannot live in presence of oxygen. 06
- 1(c) Briefly explain the environmental conditions for the growth and multiplication of bacteria. 10
- 1(d) Define percentage law of bacterial death. If a condition is so designed in which 90% of cells out of 10^6 /ml die in the first 10 minutes, then compute the total time require to sterilize the system. 09
- 2(a) Why Mycobacterium can't be stained by Gram staining procedure? 06
- 2(b) Distinguish between gram positive and gram-negative bacteria. 09
- 2(c) Briefly discuss the composition and application of the following culture media: (i) Enrichment media (ii) MacConkey's agar media 10
- 2(d) Describe the working principle of McIntosh-Fildes' anaerobic Jar with require figure. 10
- 3(a) Suppose you have been given a mixed colony agar plate cultured from the source of soaking wastewater. Now you are said to isolate specific colony from it. Write down the sequential steps of the isolation procedure. 10
- 3(b) Define bacterial spore. Briefly explain the sporulation process of bacteria. 11
- 3(c) Explain the modes of action of antimicrobial agents. 09
- 3(d) "Alcohol can act as an antimicrobial agent"--Explain. 05
- 4(a) What are biocides? Write down the applications of biocides in leather industries. 08
- 4(b) List out the microbial diseases and their resulting defects in leather. 10
- 4(c) What is immunity? Show the lines of defense of immune system of human body. 08
- 4(d) Write short notes on: (i) Sterilization (ii) Disinfection (iii) Tyndallization 09

SECTION-B

- 5(a) What is Lineweaver-Burk plot in enzyme Kinetics? State the enzyme inhibition process with the aid of Lineweaver-Burk plot. 12
- 5(b) Write down the covalent modification method of enzyme activity regulation with proper example. 08

5(c)	State the primary screening procedure for the identification of catalase and amylase producing micro-organisms.	10
5(d)	List out the factors considered in the secondary screening for sorting out the microorganism with real industrial value.	05
6(a)	What is fermentation? Write down the fermentation pathway of alcohol production from glucose.	08
6(b)	Draw the microbial growth curve. Discuss the phases of the curve where secondary metabolites are produced during fermentation.	08
6(c)	List out the unit operations of the upstream processing. Illustrate the inoculum development process.	09
6(d)	Distinguish among batch, fed-batch and continuous method of microbial fermentation.	10
7(a)	State the problem caused by fungus in the tanning industry. What are the remedies?	10
7(b)	Distinguish between the morphology of yeast and molds.	10
7(c)	Write on the symbiotic fungi with examples.	08
7(d)	Write notes on the following reproduction process of fungi: (i) Budding (ii) Sclerotia (iii) Karyogamy	09
8(a)	Write down the covalent bonding method of the immobilization of enzymes with examples.	10
8(b)	State the principles of plug flow bioreactor with the advantages and limitations.	09
8(c)	Write down the asexual spores of fungi.	10
8(d)	Write notes on the dimorphic fungi.	06

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 2nd Term Examination-2022
LE 3209
Leather Products Engineering II

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Mention the stages of leather products fabrication with examples. | 05 |
| 1(b) | What is meant by foil stamping? Describe the procedure of foil stamping. | 10 |
| 1(c) | What is meant by depth of cutting knife? Derive a relation between material thickness and depth of cutting knife. | 08 |
| 1(d) | Define skiving and splitting. Describe the skiving anatomy. | 12 |
| 2(a) | Define lining material. Construct a matrix on specific lining-specific leather products. | 08 |
| 2(b) | Which factors influence on the thread demand? | 05 |
| 2(c) | What is needle composition? Prove that thread consumption for 401 stitches, $C_s = (3L+2T)s \pm E$; where symbols have their usual meanings. | 15 |
| 2(d) | What is seam margin? Distinguish between chain stitch and lock stitch. | 07 |
| 3(a) | Define warp and weft thread with sketches. | 04 |
| 3(b) | A jacket is fabricated with a 0.9 mm thickness of leather; where seam thickness is 1.2 mm and S.P.C value is 3.0. Calculate thread consumption for lock stitch. | 08 |
| 3(c) | What are the basic weave constructions? Compare among basic weave constructions. | 12 |
| 3(d) | What is meant by poromerics? Explain-“Poromerics are known as third generation synthetics regarding texture and properties”. | 11 |
| 4(a) | What is meant by nesting of patterns? Cite the nesting principles. | 10 |
| 4(b) | Clarify the second and fourth waste during material cutting. Drive a relation to calculate second waste estimation. | 10 |
| 4(c) | Define sunk cost. Construct the general procedure of costing for leather products. | 10 |
| 4(d) | State the needle specification parameters. | 05 |

SECTION-B

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|------|---|----|
| 5(a) | Define spec sheet. Prepare a specification sheet for Blouson Jacket. | 10 |
| 5(b) | What are the ways to acquire a new product for an organization? | 04 |
| 5(c) | Briefly discuss the major steps in new product development. | 15 |
| 5(d) | “Ultimate design has an enormous impact on products service”-Explain. | 06 |

6(a)	Define theme board. Distinguish between theme board and style board.	08
6(b)	Write short notes on: (i) Motip analysis (ii) Logo design and (iii) Value addition.	12
6(c)	What is gusset? Briefly explain the following gusset preparation: (i) Three part V gusset (ii) Flat gusset.	10
6(d)	Mention the acceptance criteria for a leather bag.	05
7(a)	Define sewing. Write short notes on: (i) Feeding mechanism (ii) Unison feed and (iii) Driven roller feed.	14
7(b)	Briefly explain the following sewing complications and their possible remedies: (i) Seam puckering (ii) Skipped stitches and (iii) Needle breakage.	12
7(c)	What are the functions of the following parts of sewing machine: (i) Knee lift (ii) Check spring and (iii) Needle bar.	09
8(a)	What do you mean by zipper? Point out the different parts of a zipper.	07
8(b)	Define fusing. Write down the influencing factors of fusing materials used in leather products manufacturing.	12
8(c)	Define bag. List out the raw materials are used in bag manufacturing.	08
8(d)	Write short notes on: (i) Strike through (ii) Strike back.	08

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2022
LE 3211
Footwear Engineering II

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|--|----|
| 1(a) Define shoe last. Discuss briefly the sequence of operation for telescopic HDPE shoe last. | 12 |
| 1(b) Write down the function of the toe spring in a shoe last. Why does the toe spring change with the rigidity of the shoe sole? | 06 |
| 1(c) Distinguish between stick length and standard last length of shoe last. | 05 |
| 1(d) Write short notes on the following shoe lasts: (i) Last for Veldschoen construction (ii) American joint last and (iii) Sandal last. | 12 |
| | |
| 2(a) Provide a visual representation of the different girths of the shoe last with the explanation. | 09 |
| 2(b) Discuss briefly about the different types of shoe last grading systems. | 12 |
| 2(c) "It is mandatory to cut shoe upper materials in tight-to-toe direction in most cases"- Explain it. | 05 |
| 2(d) State the standard requirements for cutting different parts of a shoe upper. | 09 |
| | |
| 3(a) Make a comparison among different types of press cutting machines. | 12 |
| 3(b) What does the instruction sheet mean? How can an instruction sheet help determine a clicking operator's level of competence? | 08 |
| 3(c) "Shoe upper materials need to be cut under some basic principles" explain the principles for both leather and synthetic materials. | 08 |
| 3(d) Define closing. Mention the prime reasons for size marking. | 07 |
| | |
| 4(a) How can you prepare the screen for silk screen printing for stitch marking in closing department? | 09 |
| 4(b) Briefly explain the methods of punching operation. What are the points to observe while carrying out the punching operations? | 10 |
| 4(c) Write short notes on the following seams: (i) Butted seam (ii) Grain of rice seam (iii) Sprung seam and (iv) Silked seam | 16 |

SECTION-B

- | | |
|---|----|
| 5(a) Define lasting. Explain the role of elasticity and plasticity in shoe upper lasting process. | 10 |
|---|----|

5(b)	Discuss about the following shoe upper conditioning process: (i) Wetted solution (ii) Rapid conditioning cabinet and (iii) Contact mulling.	09
5(c)	Write down the common toe puff problems arise during bonding and lasting process.	10
5(d)	Sketch the following methods of lasting: (i) Slip lasting and (ii) Welted lasting.	06
6(a)	Draw a flowchart for PVC injection molding construction method of shoe making.	10
6(b)	Why vegetable tanned leather upper, insole and stiffener can't be used during direct vulcanizing process of molded construction?	06
6(c)	Explain the mechanism of rubber vulcanization in molded construction with specific reaction and figure.	12
6(d)	Why PU sole is considered as one of the best soling materials?	07
7(a)	Write down the criteria of different shoe upper preparation process during cemented construction technique.	06
7(b)	Discuss about different chemical treatment processes that are used during shoe sole preparation in cemented construction.	09
7(c)	Explain the sequence of hand lasting method of shoe with figure.	15
7(d)	Write down the advantages of machine welted construction process.	05
8(a)	"Insole attachment varies among different shoe construction"-Justify it.	10
8(b)	Explain the following terms of attaching operation during welted construction: (i) Inseam trimming (ii) Welt beating.	08
8(c)	Mention different safety problems arise during PU injection molding process.	10
8(d)	Why adhesive is applied in two way methods during shoe injection molding process?	07

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 2nd Term Examination-2022
ME 3221
Manufacturing Process

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

- | | |
|--|----|
| 1(a) In what circumstances casting is preferable manufacturing method compared to machining? | 05 |
| 1(b) Draw a complete mold and level it properly. | 10 |
| 1(c) Explain the properties of molding sand in detail. | 12 |
| 1(d) Why shaking allowance is called negative allowance? Explain. | 08 |
| | |
| 2(a) What is die casting? Classify it. | 05 |
| 2(b) Explain the submerged plunger type die casting process. | 15 |
| 2(c) With the help of diagram, briefly explain lost wax casting method. | 10 |
| 2(d) What types of defects are generally found in casting? | 05 |
| | |
| 3(a) How gases are mixed in a gas welding torch? What are the functions of gas mixer? | 08 |
| 3(b) Explain seam welding process with neat sketches and mention its advantages, limitations and applications. | 15 |
| 3(c) Describe the characteristics of oxy-acetylene gas welding frames with figure. | 12 |
| | |
| 4(a) What are the elements used in the coating of Arc welding electrode? Explain their functions. | 12 |
| 4(b) Compare and contrast between welding and brazing. | 10 |
| 4(c) What is recrystallization? Write down the common purposes of cold working process. | 06 |
| 4(d) Write a short note on Extrusion. | 07 |

SECTION-B

- | | |
|---|----|
| 5(a) What is manufacturing process? Explain how the knowledge of manufacturing process can be used in leather and footwear industries. | 08 |
| 5(b) Describe the principle of metal cutting. Identify different types of chips formed in metal cutting process. Why chip breaker is used? | 10 |
| 5(c) Draw a single point cutting tool and show its different angles and elements. | 08 |
| 5(d) Explain the following terms: (i) Cutting speed, (ii) Feed, and (iii) Depth of cut | 09 |
| | |
| 6(a) Define chip thickness ratio. Show that, $\tan \beta = \frac{r \cos \alpha}{1 - r \sin \alpha}$, where symbols have their usual meaning. | 12 |

6(b)	Illustrate the relationship between cutting speed and tool life.	07
6(c)	Define cutting fluid. State the functions and properties of cutting fluid.	10
6(d)	What is machine tool? Give example. Write down the functions of machine tool.	06
7(a)	Explain the machine working principle of a lathe machine.	10
7(b)	Differentiate drilling, boring and reaming from each other.	05
7(c)	Why universal milling machine is named as universal? Describe the lever rocker arm quick return mechanism for shaper machine.	08
7(d)	What is indexing? Describe simple and angular indexing method.	12
8(a)	Write short note on: (i) Honing, (ii) Lapping and (iii) Superfinishing	06
8(b)	Explain centerless grinding method with its advantages and limitations.	10
8(c)	What are the various methods of making screw threads? Describe any one of them.	07
8(d)	Explain the EDM process and mention its advantages and limitations.	12

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2022
ME 3219
Fluid Mechanics and Machineries

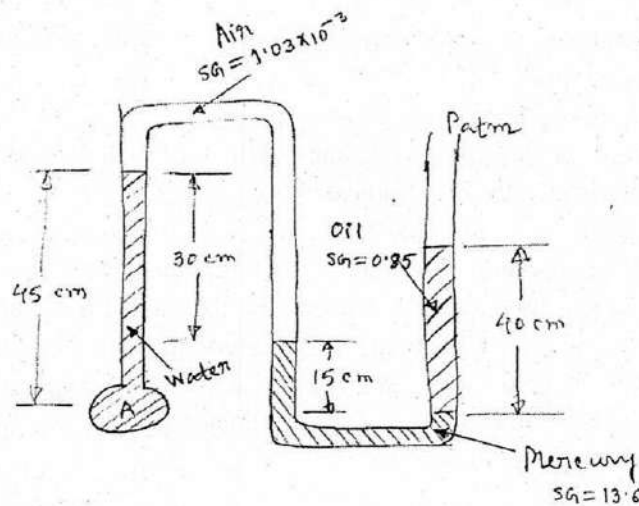
Time: 3.0 Hours

Full Marks: 210

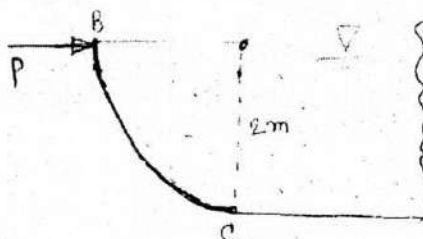
- N.B. i) Answer any **THREE** questions from each section in separate scripts.
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SECTION-A

- 1(a) What is mean by 'fluid' and 'fluid particle'? Why do the 'fluid as continuum' assumption is important in classical fluid mechanics? Explain with sketch. 10
- 1(b) Why a droplet of water takes spherical shape? Why the droplet will break out if more mass of water added to it? 08
- 1(c) Explain why the viscosity of liquid decreases with temperature but that of gas increases with temperature? 07
- 1(d) A solid cylindrical needle of diameter d , length L and density ρ may float on a liquid surface. Neglect buoyancy and assume contact angle of 0° . Derive expression for the maximum diameter of the needle that will be able to float on the surface. Calculate the diameter for a steel needle (SG = 7.84) in water. Take $\rho_s = 0.07 \text{ N/m}$. 10
- 2(a) Show that pressure into the static fluid remains constant along the horizontal surface but varies along the vertical direction. 10
- 2(b) Explain how to measure the atmospheric pressure using barometer? 07
- 2(c) Show graphically the relationship between gage pressure, absolute pressure and atmospheric pressure and vacuum pressure. 06
- 2(d) Determine the gage pressure at point A in Pa as in following Figure. Is the gage pressure higher or lower than atmospheric? 12



- 3(a) How the resultant hydrostatic force on submerged curved surface can be determined? Explain with neat sketches. 10
- 3(b) Show that the resultant force on a body, submerged in a liquid does not pass through the centroid but rather always below it, if the surface is non-horizontal. 15
- 3(c) The quarter circle gate BC is hinged at C. Find the horizontal force P required to hold the gate stationary. Neglect the weight of the gate (see the following figure): 10

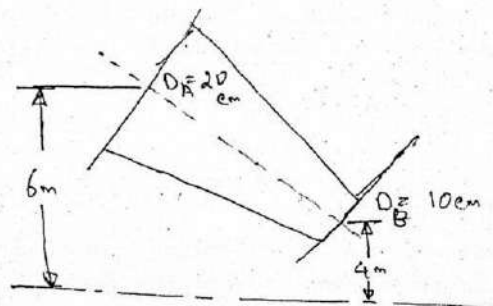


- 4(a) Why the direction of buoyant force is vertically upward? Show that buoyant force is independent of the distance of the body from the free surface. 08

- 4(b) How metacentric height is a measure of stability for floating object? With neat sketch demonstrate the stability and instability conditions for floating object. 12
- 4(c) A solid cylinder of diameter 4.0 m has a height of 4.0 m. If the specific gravity of the material of the cylinder is 0.6 and it is floating in water with its axis vertically, (i) Find the metacentric height of the cylinder and (ii) Find the buoyancy force acting on the cylinder. State whether the equilibrium is stable or unstable. 15

SECTION-B

- 5(a) Explain the methods of visualization of fluid flow with appropriate diagrams. Deduce the equation of streamline. 10
- 5(b) Derive the extended form of Bernoulli's equation. Provide a brief explanation of its constituent terms. 12
- 5(c) The water flows through a pipe having diameters of 20 cm at section (A) and 10 cm at section (B). The distance of section (A) and (B) from the datum is provided in the following figure. If the pressure difference between section (A) and (B) is 10 kPa, determine (i) flow rate and (ii) flow velocity at both sections. 13



- 6(a) Demonstrate that the incorporation of air vessels has the potential of significantly reducing the work required against friction. 10
- 6(b) Draw the characteristic curves of a reciprocating pump. 03
- 6(c) Describe the change in indicator diagram when acceleration and frictional effects are considered, in comparison to the ideal indication diagram. 08
- 6(d) A single acting reciprocation pump has a cylinder of 100 mm diameter and a stroke length of 200 mm. The center of the pump is 3 m above the water level in the sump and 20 mm below the water level in a tank to which water is delivered by the pump. The diameter and length of the suction pipe are 50 mm and 5 m while the delivery pipe 40 mm and 30 m, respectively. Determine the maximum speed at which the pump may be run without separation, if separation occur at 3.8 m of water (absolute). If an air vessel is incorporated, what would be the change in maximum speed? Consider $f=0.036$. 14
- 7(a) What is a negative slip in a reciprocating pump and when does it appear? 06
- 7(b) Explain the main components of a centrifugal pump with necessary sketches. Describe the working principle of a centrifugal pump with appropriate diagram. 16
- 7(c) What are the differences between reciprocating pumps and centrifugal pumps? For lifting water to a 4th floor building which type of pump would you choose? Justify your choice. 13
- 8(a) Why is the use of differential approaches in solving fluid dynamics problems necessary, despite the simplicity of the integral approach? 07
- 8(b) What is an air compressor and where are they used? Classify compressors. 10
- 8(c) Derive the general form of Reynolds transport theorem. Hence deduce the continuity equation using the theorem. 13
- 8(d) Distinguish among fan, blower and compressor. 05